

**FINAL ENVIRONMENTAL PROJECT BRIEF REPORT
FOR THE PROPOSED ARTISAN MINING PROJECT
UNDER ARTISAN MINERAL RIGHT NO. 37369-HQ-
AMR IN MANDELE VILLAGE OF CHIEF SIKONGO'S
AREA IN CHIRUNDU DISTRICT BY MULONGA
MINING COOPERATIVE SOCIETY LIMITED.**



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EXECUTIVE SUMMARY

Background

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Mulonga Mining Cooperative Society Limited was registered as a co-operative society under the Cooperative Societies Act on the 30th of January, 2024 to help local communities explore and utilize the mineral resources in their areas thereby providing a platform for local people to benefit from the mineral wealth of the country.

The Cooperative intends to engage in artisanal mining activities of mainly copper under Artisanal Mining Rights No. 37369 HQ-AMR in Mandele area of Chirundu District. As a requirement under the provisions of the Environmental Management Act No. 12 of 2011 and the Environmental Impact Assessment Regulations SI No. 28 of 1997, a project of this nature requires an Environmental and Social Impact Assessment (ESIA) to be conducted before implementation. It is against this background that Mulonga Cooperative Society Limited engaged Elizkoffi Consult Limited to undertake the ESIA study on their behalf.

Project Description

Location

The proposed site is located in Mandale area under Chief Sikongo in Chirundu District of Southern Province. The site can be accessed using the Chirundu – Lusaka (T2) road and is about 5.45Km off the T2 road and about 24Km from Chirundu - Siavonga junction. Notable landmark is DPT towing company which is about 8.78 km South-East of the proposed mining site. The site lies on GPS coordinates 16°00'06.00S" and 28°35'12.00E".

Nature

The proponent applied for artisanal mineral rights for Citrine, Copper, Feldspar, Gold, Gypsum, Anhydrite, Lead, Manganese, Mica, Nickel and Zinc. Mulonga Cooperative wishes to set up an artisanal mine that will be mining for aforementioned ores using an open pit method on a 6.58Ha piece of land. The mine which is expected to produce about 100 tons of copper per day will comprise the following facilities:

- Worker's quarters with kitchen measuring 200m²;
- A store room measuring 100 m²;
- A stockpiling and loading area measuring 1500 m²
- Ablution block measuring 10 m²
- Waste rock area measuring 1500 m²

Mulonga Mining Cooperative Society Limited will undertake the following activities,

- Exploration and sampling activities
- Clearing vegetation for road and mining facilities
- Open pit mining;
- Stock piling and loading of mineral ore

- Transportation of mineral ore
- Management of waste rock
- Occasional blasting depending on underlying rock type
- Dust suppression

Project Objective

To undertake mining (Citrine, Copper, Feldspar, Gold, Gypsum, Anhydrite, Lead, Manganese, Mica, Nickel and Zinc) activities that will take advantage of the mineral wealth of the area for economic gain.

Shareholders

The cooperative is wholly owned by registered community members thereby has no shareholders.

Project Implementation Date and Cost of the Project

The project is scheduled to be implemented as soon as ZEMA grants approval.

The total cost of the project is **K500, 000-00**.

The Legislative and Regulatory Frameworks

The governing legislative instruments on this proposed project comprise but not limited to the following:

- i) Environmental Management (Amendment) Act No. 8 of 2024 as read together with the Environmental Management No. 12 of 2011
- ii) The Environmental Impact Assessment (EIA) Regulations, SI 28 of 1997
- iii) Environmental Management (Licensing) Regulations, SI No. 112 of 2013
- iv) The Water Resources Management Act, No. 21 of 2011
- v) The Forest Act No. 4 of 2015
- vi) The Zambia Wildlife Act No. 14 of 2015
- vii) The National Heritage Conservation Act (CAP 173)

- viii) Public Health Act Cap 295 of 1978 Part V Section 31
- ix) The Local Government Act No. 2 Of 2019 Section 16
- x) The Mines and Minerals Development (Amendment) No. 14 of 2016
- xi) The Road Traffic Act No. 11 of 2002
- xii) The Road Traffic Act No. 11 of 2002

The international conventions and agreements that Zambia is party to and are relevant to the project include:

- i) United Nations Framework Convention on Climate Change Ratified 1997 UN-FCCC
- ii) Convention on International Trade in Endangered Species of Wild Fauna and Flora Ratified: 1975
- iii) UN Convention on Biological Diversity (1992)
- iv) Convention Concerning the Protection of Workers Against Occupational Hazards in the Working Environment Due to Air Pollution

Project Alternatives

The alternatives considered included;

I. Project Location

Options considered included developing the said project at the proposed site or finding another site. The proponent only has one site which they have purchased and is on title hence constructing at the proposed was the chosen option.

II. Power Source

The proponent considered solar power, connecting to the national grid and using a diesel-powered generator. Solar and diesel-powered generator were the chosen options.

III. No Project Option

This option would mean foregoing the mineral wealth of the area and the benefits it could bring to the community. This option was not considered as implementation of the project would bring high economic benefits to the area and surrounding communities.

IV. Sewage Management

Two options of sewage management were considered for the project. This included use of septic tanks and installation of mobile latrines. Use of septic tank was the chosen option.

Project Potential Impacts

The proposed project has potential environmental and social impacts. The

following are some of the impacts that were identified during the ESIA study:

- **Impact on Land and Soil:** Soil erosion impacts can be caused by site preparation and construction activities (vegetation clearance, earth works, etc.).
- **Impact on Biodiversity and Vegetation:** Loss of vegetation and wildlife habitats due to preparation/construction activities which require vegetation clearance.
- **Waste Generation and Management:** Pollution from both hazardous and non-hazardous wastes
- **Impact on Air Quality:** Dust emissions mining and transportation activities and emissions from vehicles.
- **Impact of Noise:** machinery for construction and mining as well as vehicles during operation will generate noise.
- **Community Health and Safety:** The presence of the mine has the potential to affect community health such as increased transmission of communicable diseases – due to in-migration.
- **Impacts to worker health and safety and rights in relation to employment:** Workers' rights including occupational health and safety will need to be considered
- **Human- animal Conflict:** the area is a greenfield which potentially has snakes.

Mitigation Measures

Many of the impacts mentioned above can be mitigated by thoroughly implementing and monitoring the measures outlined in the Environmental and Social Management Plan (ESMP), with no or only minor negative impacts remaining.

Impact on Biodiversity and Vegetation: Restrict clearing of vegetation only to relevant areas

Impact on Land and Soil: All onsite sewage leakages will be monitored to avoid land and soil contamination. All spillages will be managed according to local regulations. Mining methods will be supervised by an experienced mining personnel.

Impact on Air Quality: Watering of the area and surroundings during the construction stage will be undertaken regularly to assure dust suppression and proper personal protective equipment will be provided to site workers. Speed limits will also be adhered to avoid dust generation.

Waste Generation and Management: solid waste will be collected and disposed of according to the existing regulations of solid waste disposal approved by the local authority.

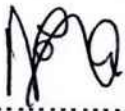
Impact of Noise: All mobile vehicles and equipment will have noise reducers and will be properly serviced and construction and mining operation will be limited to day hours only.

Human-Animal Conflict: the proponent will adhere strictly to the DNPW guidelines.

Community Health and Safety: Employing of local residents to minimize the spread of communicable diseases and conducting sensitization talks. Further condoms and masks to be provided to all workers.

Conclusion

Mulonga Mining Cooperative Society Limited plans to develop and implement an effective Environmental Management Plan to address all the projected negative impacts as well as develop and implement an emergency preparedness and response plan that will cover the anticipated emergency situations.



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DIRECTOR

Mulonga Mining Cooperative Society Limited

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LIST OF ACRONYMS AND ABBREVIATIONS

AIDS -	Acquired Immunodeficiency Syndrome
CBD -	Convention on Biological Diversity
EPB -	Environmental Project Brief
ESIA -	Environmental and Social Impact Assessment
ECZ -	Environmental Council of Zambia
EMA -	Environmental Management Act
EMMP -	Environmental Management and Monitoring Plan
EPPCA -	Environmental Protection and Pollution Control Act
GRZ -	Government of the Republic of Zambia
Ha -	Hectares
HIV -	Human Immuno-deficiency Virus
DNPW-	Department of National Park and Wildlife
NEAP -	National Environmental Action Plan
NCS -	National Conservation Strategy
TORs -	Terms of Reference
WARMA -	Water Resources Management Authority
ZEMA -	Zambia Environmental Management Agency

1.0 INTRODUCTION

Mulonga Mining Cooperative Society Limited was registered as a co-operative society under the Cooperative Societies Act on the 30th of January, 2024 to help local communities explore and utilize the mineral resources in their areas thereby providing a platform for local people to benefit from the mineral wealth of the country.

The Cooperative intends to engage in artisanal mining activities of mainly copper under Artisanal Mining Rights No. 37369 – HQ -AMR In Mandale area of Chirundu District. As a requirement under the provisions of the Environmental Management Act No. 12 of 2011 and the Environmental Impact Assessment Regulations SI No. 28 of 1997, a project of this nature requires an Environmental and Social Impact Assessment (ESIA) to be conducted before implementation. It is against this background that Mulonga Cooperative Society Limited engaged Elizkoffi Consult Limited to undertake the ESIA study on their behalf.

1.1 Summary description of the project

Nature

The proponent applied for artisanal mineral rights for Citrine, Copper, Feldspar, Gold, Gypsum, Anhydrite, Lead, Manganese, Mica, Nickel and Zinc. Mulonga Cooperative wishes to set up an artisanal mine that will be mining for aforementioned ores using an open pit method on a 6.58Ha piece of land. The mine which is expected to produce about 100 tons of copper per day will comprise the following facilities:

- Worker's quarters with kitchen measuring 200m²;
- A store room measuring 100 m²;
- A stockpiling and loading area measuring 1500 m²;
- Ablution block measuring 10 m² and;
- Waste rock area measuring 1500 m².

Mulonga Mining Cooperative Society Limited will undertake the following activities,

- Exploration and sampling activities
- Clearing vegetation for road and mining facilities
- Open pit mining;
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- Transportation of mineral ore
- Management of waste rock
- Occasional blasting depending on underlying rock type
- Dust suppression

1.2 Objective of the project

To undertake mining (Citrine, Copper, Feldspar, Gold, Gypsum, Anhydrite, Lead, Manganese, Mica, Nickel and Zinc) activities that will take advantage of the mineral wealth of the area for economic gain.

1.3 Brief description of the location

Project Description

Location

The proposed site is located in Mandale area under Chief Sikongo in Chirundu District of Southern Province. The site can be accessed using the Chirundu – Lusaka (T2) road and is about 5.45Km off the T2 road and about 24Km from Chirundu - Siavonga junction. Notable landmark is DPT towing company which is about 8.78 km South-East of the proposed mining site. The site lies on GPS coordinates 16°00'06.00S" and 28°35'12.00E".

Table 1: coordinates for project site

Corner points	Latitude Degrees (S)	Minutes	Seconds	Longitude Degrees	Minutes	Seconds
1	16	00	06	28	35	12.00
2	16	00	06	28	35	24.00
3	16	00	12	28	35	24.00
4	16	00	12	28	35	12.00

1.4 Particulars of Shareholders/Directors

The project is wholly owned by registered community members of Mandela area.

1.5 The developer's physical address and the contact person

Developers details

Cooperative : Mulonga Mining Cooperative Society Limited

Address: C/O Siagweemu Primary School, P.O Box 21, Chief Sikongo area, Mandale Village, Chirundu.

Tel: +260977310941

Email: ndalamap7007.pn@gmail.com

Contact Person: Mr Patt Ndalama

Address: 51 Kubu Street, Libala stg 3, Lusaka

Designation: Secretary

Tel: +260977310941

Email: ndalamap7007.pn@gmail.com

1.6 Track Record/Previous Experience of Enterprise Elsewhere

Mulonga Mining Cooperative Society Limited is a newly incorporated Cooperative but the leading members have over 5 years' experience in the mining field.

1.7 Project Cost Investment and Proposed Project Implementation Date

The project is scheduled to be implemented as soon as ZEMA issues approval. The total cost of the project is projected at **K500,000**.

1.8 Scope of Study and Methodology

The Environmental and Social Impact study adheres to the requirements of the Environmental Impact Assessment (EIA) regulations of 1997. The EIA regulations demand that socio - economic and environmental impacts consequential to the implementation and operation of the project are assessed well in advance. Elizkoffi Consult Limited commissioned the development of an Environmental and Social Impact Assessment (ESIA) to assess the baseline, environmental trends and socio-economic data of the project area, identify significant environmental and socio-economic impacts of the project and propose possible mitigation measures.

This ESIA study was undertaken through desk review of the relevant literature and specialized study of the various components of the project i.e. water, air, soil, noise, ecology, flora, fauna, climate and geology. Other studies included socio-economic and cultural related studies

2.0 POLICY, LEGAL AND INSTITUTIONAL FRAMEWORK

This section will cite all policy, legal and institutional framework relevant to the project

2.1 Environmental Policy Framework

Zambia launched its National Policy on Environment in 2007. The environmental policy document identifies deforestation, land degradation, wildlife depletion, soil erosion, loss of land productivity, inadequate sanitation and air and water pollution as the main environmental challenges the country is facing.

According to the Ministry responsible for the Environment, the main purpose of the National Policy on Environment is to “create a comprehensive framework for effective natural resource utilization and environmental conservation which will be sensitive to the demands of sustainable development. The Policy calls upon all institutions, Non-Governmental Organizations (NGOs), and community based or people’s organizations whose activities affect the environment in any way to carry out their activities judiciously in order to maintain the productivity and integrity of the environment.

The Policy recognizes the need for ESIA studies in all development projects in order to eliminate or mitigate adverse environmental impacts and enhance the benefits of the projects. The National Policy on Environment states that “EIAs will be required as deemed necessary to ensure that public and private sector development options are environmentally sound and sustainable and that any environmental consequences are recognized early and taken into account during project design and implement.

2.2 Regulatory Framework

Due to the need for balancing environmental requirements, economic activities and social needs, the Government of the Republic of Zambia (GRZ) adopted the National Conservation Strategy (NCS) in 1985. This was upgraded to the National Environmental Action Plan (NEAP) in 1992 with the same aim of fostering sustainable development. The NCS and NEAP are the foundation pillars of environmental laws in Zambia. The NCS facilitated development of the Environmental Protection and Pollution Control Act (EPPCA) in 1990 which also facilitated formulation of the Environmental Council of Zambia (ECZ) in 1992. The EPPCA was repealed in 2011 into the Environmental Management Act No. 12 of 2011 which was amended to the Environmental Management Act No. 8 of 2024, which is the current supreme environmental management law. Consequently, the name of the mother environmental regulatory body changed from ECZ to Zambia Environmental Management Agency (ZEMA)

2.2.1 Environmental Management (Amendment) Act (EMA) No. 8 of 2024 as read together with the EMA No. 12 of 2011

This is the principal legislation governing environmental management in Zambia. The Act provides for the continuation and renaming of the Environmental Council of Zambia (ECZ) as ZEMA. The Agency is mandated to ensure the sustainable management of natural resources and protection of the environment, and the prevention and control of pollution. In relation to this project, some of the functions of ZEMA are to review environmental impact assessment reports, and undertake environmental auditing and monitoring. The Act also provides for public participation in environmental decision-making and access to environmental information. In particular, section 29 of the Act states that "A person shall not undertake any project that may have an effect on the environment without the written approval of the Agency, and except in accordance with any conditions imposed in that approval".

Relevance: The project represents activities which if unchecked can have negative effects on the environment, and hence will need to comply with the EMA of 2011 and 2024.

Compliance thereof: The proponent will ensure the project obtains all necessary approvals as prescribed by the EMA of 2011, and abides by the prescriptions of the law.

2.2.2 The Environmental Impact Assessment Regulations, SI 28 of 1997

The Environmental Impact Assessment (EIA) Regulations, Statutory Instrument 28 of 1997, Part II, under the Environmental Protection and Pollution Control Act of 1990 demands that before a developer commences implementing a project, an EIA be prepared and submitted to the relevant regulatory authority for review and approval.

Relevance: development of mine falls under a project category requiring the preparation of an EPB and an EMP containing a description of adverse impact mitigation measures and monitoring programs to be implemented.

Compliance thereof: This EPB has been prepared in compliance with the above legal provision.

2.2.3 Environmental Management Licensing Regulations (SI No 112 of 2013)

This Regulation provides the main framework under which Licensing by ZEMA is conducted for various activities.

Relevance: Some activities might lead to generation of hazardous waste and discharge of effluent

Compliance thereof: the proponent will obtain the required licenses and ensure to comply with the conditions

2.2.4 Workers Compensation Act, 1999

An Act to revise the law relating to the compensation of workers for disabilities suffered or diseases contracted during the course of employment; to provide for the merger of the functions of the Workers' Compensation Fund Control Board and the Pneumoconiosis compensation fund for the compensation of workers disabled by

accident occurring, or diseases contracted in the course of employment; to provide for the payment of compensation to dependents of workers who die as a result of accidents or diseases; to employers; to provide for the appointment and powers of a Workers' Compensation Commissioner, the establishment and functions of a workers' Compensation Fund Board and workers Compensation Tribunal; and to provide for matters connected with and incidental to the foregoing.

Relevance: The proponent will employ both skilled and unskilled workers for various positions.

Compliance thereof: The proponent will make mandatory annual contribution to the Workers Compensation Fund Control Board to cover the workers should they be injured, develop a terminal illness or in the unfortunate event of a fatality at the work place.

2.2.5 The Public Health Act Cap 295 of 1996

The Act provides for and regulates all matters connected with public health in the country with the Local Authority of each District as the enforcement agency.

Relevance: The proposed project will handle food and also generate various forms of waste.

Compliance thereof: The EIA provide for stringent air, water and sanitation systems to be employed at the project sites. Further to this, all guidelines provided by the local authority will be adhered to.

2.2.6 Water Resources Management Act No. 21 of 2011

This repeal and replaces the Water Act, 1949. It establishes Water Resources Management Authority and its functions and powers; provides for the management, development, conservation, protection and preservation of the water resource and its ecosystems; provide for the equitable, reasonable and sustainable utilization of the water resource; create an enabling environment for adaptation to climate change; provide for the constitution, functions and composition of catchment councils, sub-catchment councils and water users associations; provide for international and regional cooperation in, and equitable and sustainable utilization of, shared water resources; provide for the domestication and implementation of the basic principles and rules of international law relating to the environment and shared water resources as specified in the treaties, conventions and agreements to which Zambia is a State Party.

Relevance: project activities will involve drilling of a borehole on site

Compliance thereof: The proponent will ensure that all relevant permits are obtained before engaging in any regulated activity.

2.2.7 Employment Code Act No. 3 of 2019

This Act provides conditions under which employees should work in Zambia. The Act covers both temporary and permanent employees. Generally, this Act talks about employee protection and social security requirements. Major provisions include:

- (a) Minimum contractual age;
- (b) Establishment of employment contracts;

- (c) Settlement of disputes arising from such contracts of employment;
- (d) The appointment of Labour Officers and other staff for the administration of the Act; And
- (e) Certain conditions of employment such as ordinary leave, sick leave, maternity, redundancy and welfare of employees.

Relevance: The Proponent will have employees during the three stages of the project.

Compliance thereof: The proponent will comply with this Act by employing workers of the right ages, establishing contracts, conditions of service and settling disputes in accordance with this act.

2.2.8 The Urban and Regional Planning Act No. 3 of 2015

An Act to provide for development, planning and administration principles, standards and requirements for urban and regional planning processes and systems; provide for a framework for administering and managing urban and regional planning for the Republic; provide for a planning framework, guidelines, systems and processes for urban and regional planning for the Republic; establish a democratic, accountable, transparent, participatory and inclusive process for urban and regional planning that allows for involvement of communities, private sector, interest groups and other stakeholders in the planning, implementation and operation of human settlement development; ensure functional efficiency and socio-economic integration by providing for integration of activities, uses and facilities; establish procedures for integrated urban and regional planning in a devolved system of governance so as to ensure multi-sector cooperation, coordination and involvement of different levels of ministries, provincial administration, local authorities, traditional leaders and other stakeholders in urban and regional planning; ensure sustainable urban and rural development by promoting environmental, social and economic sustainability in development initiatives and controls at all levels of urban and regional planning; ensure uniformity of law and policy with respect to urban and regional planning; repeal the Town and Country Planning Act, 1962, and the Housing (Statutory and Improvement Areas) Act, 1975; and provide for matters connected with, or incidental to, the foregoing.

Relevance: The Act is relevant because even if the project is based in traditional land, the guidelines under this act will be adhered to ensure that the development in the area are sustainable.

Compliance thereof: Mulonga Mining Cooperative Society Limited shall comply with the conditions set by the local authority and will ensure that all necessary approvals are obtained.

2.2.9 The Zambia Wildlife Act No. 14 of 2015

An Act to provide for the winding up of the affairs of the Zambia Wildlife Authority; establish the Department of National Parks and Wildlife in the Ministry responsible for tourism; provide for the appointment of a Director and other officers responsible for National Parks and Wildlife;

Provide for the transfer of the functions of the Authority to the Ministry responsible for tourism, Department of National Parks and Wildlife and Director of National Parks and Wildlife; establish the Wildlife Management Licencing Committee; provide

for the establishment, control and management of National Parks, bird and wildlife sanctuaries and for the conservation and enhancement of wildlife eco-systems, biological diversity and objects of aesthetic, pre-historic, historical, geological, archaeological and scientific interest in National Parks; provide for the promotion of opportunities for the equitable and sustainable use of the special qualities of public wildlife estates; provide for the establishment, control and co-management of Community Partnership Parks for the conservation and restoration of ecological structures for non-consumptive forms of recreation and environmental education; provide for the sustainable use of wildlife and the effective management of the wildlife habitat in Game Management Areas; enhance the benefits of Game Management Areas to local communities and wildlife; involve local communities in the management of Game Management Areas; provide for the development and implementation of management plans; provide for the regulation of game ranching; provide for the licensing of hunting and control of the processing, sale, import and export of wild animals and trophies; provide for the implementation of the Convention on International Trade in Endangered Species of Wild Fauna and Flora, the Convention on Wetlands of International Importance especially as Waterfowl Habitat, the Convention on Biological Diversity, the Lusaka Agreement on Cooperative Enforcement Operations Directed at Illegal Trade in Wild Fauna and Flora and other international instruments to which Zambia is party; repeal the Zambia Wildlife Act, 1998; and provide for matters connected with, or incidental to, the foregoing.

Relevance: The project site is in a greenfield area which is habitat to wild animals such as snakes and monkeys.

Compliance there of: The proponent will follow the requirements of this Act and the provisions of the DNPW approval guidelines.

2.2.10 National Council for Construction Act No. 10 of 2020

This is an act to provide for the establishment of the national Council for construction and to define its function; to provide for the promotion and development of the construction industrial in Zambia; to provide for the registration of the contractors to provide for the affiliation to the council of professional bodies or organization whose members are engaged in activities related to the construction industry; to provide for the regulation of the construction industry; to provide for the regulation of the construction industrial; to provide for the establishment for the construction school; to provide for the training of persons engaged in construction or in activities related to the construction; and provide for matters connected with or incidental to the fore-going.

Relevance: The project involves the construction of a number of facilities

Compliance there of: The proponent will comply with the requirements of the act by hiring a contractor who is registered with the NCC.

2.2.11 Lands Act Cap 184 of 1995

The Land Act of 1995 was enacted to guarantee peoples' right to land while enhancing development. The Act recognizes the holding of land under customary tenure and the Chief's role has been legally recognized, such that land cannot be converted or

for the establishment, control and management of National Parks, bird and wildlife sanctuaries and for the conservation and enhancement of wildlife eco-systems, biological diversity and objects of aesthetic, pre-historic, historical, geological, archaeological and scientific interest in National Parks; provide for the promotion of opportunities for the equitable and sustainable use of the special qualities of public wildlife estates; provide for the establishment, control and co-management of Community Partnership Parks for the conservation and restoration of ecological structures for non-consumptive forms of recreation and environmental education; provide for the sustainable use of wildlife and the effective management of the wildlife habitat in Game Management Areas; enhance the benefits of Game Management Areas to local communities and wildlife; involve local communities in the management of Game Management Areas; provide for the development and implementation of management plans; provide for the regulation of game ranching; provide for the licensing of hunting and control of the processing, sale, import and export of wild animals and trophies; provide for the implementation of the Convention on International Trade in Endangered Species of Wild Fauna and Flora, the Convention on Wetlands of International Importance especially as Waterfowl Habitat, the Convention on Biological Diversity, the Lusaka Agreement on Cooperative Enforcement Operations Directed at Illegal Trade in Wild Fauna and Flora and other international instruments to which Zambia is party; repeal the Zambia Wildlife Act, 1998; and provide for matters connected with, or incidental to, the foregoing.

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Relevance: The project involves the construction of a number of facilities **Compliance there of:** The proponent will comply with the requirements of the act by hiring a contractor who is registered with the NCC.

2.2.11 Lands Act Cap 184 of 1995

The Land Act of 1995 was enacted to guarantee peoples' right to land while enhancing development. The Act recognizes the holding of land under customary tenure and the Chief's role has been legally recognized, such that land cannot be converted or

alienated without approval of the chief.

Relevance: The project site falls under customary land which is under the administration of Chief Sikongo.

Compliance there of: Mulonga Mining Cooperative Society Limited will ensure to work hand in hand with the local leadership and obtain all necessary consents where necessary.

2.2.12 The Occupational Health and Safety Act No.36 of 2010

Provides for preventing work related injuries, illnesses, and death by issuing and enforcing workplace health and safety standards. Hundreds of OHS safety standards, covering everything from first aid to fire protection, apply to most federal and private workplaces. OHS allows employees to question unsafe conditions, request inspections, and file lawsuits to correct unsafe conditions without retaliation.

Relevance: Construction and operation activities will expose workers to occupational and health risks.

Compliance there of: The developer will come up with programs to educate its workers on workplace safety, health and environmental best practices. They will also provide Personal Protective Equipment to its employees.

2.2.13 National Heritage Conservation Commission Act CAP 173 of 1989

The National Heritage Conservation Commission Act CAP 173 of 1989 stipulates preservation and protection of ancient cultural and natural heritage resources and objects of aesthetic, historical and archaeological value. In this Act, "Ancient Heritage is defined as being among other things, any structure, settlement previously inhabited, land mark, burial place or any other item designated by the commission which is known or believed to have been erected, constructed or used before 1st January 1924. The Act also provides for the formation of the National Heritage and Conservation Commission which is the responsible institution.

Relevance: The project will involve earth movement works in the construction phase hence disturbance of virgin areas that might have historical significance.

Compliance there of: This developer will report any such existence and will preserve and protect such if found on site.

2.2.14 The Standards Act No. 4 of 2017

An Act to continue the existence of the Zambia Bureau of Standards and re-define its powers and functions; provide for standardization and quality assurance of products and services through the setting of national standards and provision of conformity assessment services for products and services; repeal the Standards Act, 1994; and provide for matters connected with, or incidental to, the foregoing.

Relevance: The development of the camp and its auxiliary facilities will require procurement of construction materials

Compliance there of: All materials to be used will meet the standards of the bureau.

2.2.15 The Public Health (infected Areas) Coronavirus Disease) Regulations, 2020, Statutory Instruments 21 and 22

Statutory Instrument 21 of 2020 designates COVID-19 as a notable disease and Statutory Instrument 22 provides additional management and control of COVID 19. These S.Is provide for mandatory Screening and quarantine of travelers and international travelers, restriction of foreign travels, hygiene practices to be conducted at workplaces, public and schools, self-quarantine of travelers coming in the country, social distancing and mandatory wearing of facemasks

Relevance: the project will involve interaction of local and international personnel.

Compliance thereof: The developer will adhere to the provisions on the regulations through provisions that will be relevant at any given time.

2.3 The National HIV/AIDS/STI/TB Act of 2002 Section 4

The Act provides for the establishment of the National HIV/AIDS/STI/TB council and provides for the support to the development, monitoring and evaluation of multi-sectorial and economic impacts of HIV, STI, AIDS, and TB.

Relevance: influx of people seeking employment and casual interactions between locals and tourists.

Compliance: Awareness programs will be arranged for all the workers as well as the distribution of condoms in the convenient places.

2.3.1 Solid Waste Management Act No. 20 of 2018

An Act to provide for the sustainable regulation and management of solid waste; general and self-service solid waste services; the incorporation of solid waste management companies and define their statutory functions; the licensing and functions of solid waste service providers, operators and self-service solid waste providers and provide for their functions; the regulation, operation, maintenance and construction of landfills and other disposal facilities; the setting and approval of tariffs for management of solid waste and provision of solid waste services; and matters connected with, or incidental to, the foregoing.

Relevance: Activities will produce solid waste that if not properly handled can disturb public health and also contribute to human-animal conflict

Compliance thereof: The developer will store, manage, transport and dispose solid waste in accordance with the guidelines of this act.

2.3.2 Local Government Act No. 2 of 2019 section 16

An Act to provide for an integrated local government system; give effect to the decentralization of functions, responsibilities and services at all levels of local government; ensure democratic participation in, and control of, decision making by the people at the local level; revise the functions of local authorities; provide for the review of tariffs, charges and fees within the area of a local authority; provide for the proceedings of the council and committees; provide for the role of traditional leadership in democratic governance and provide for matters connected with, or incidental to, the foregoing

Section 16: A local authority shall discharge functions conferred on it by this Act within the area of that local authority, and may, with the approval of the Minister discharge that function outside the area of that local authority.

Relevance: the project site being in the council area, the facility will require local licensing e.g. fire certificates.

Compliance: All the required certificates will be furthered at the Chirundu Town Council.

2.3.3 The Forest Act No. 4 of 2015

provide for the conservation and use of forests and trees for the sustainable management of forests ecosystems and biological diversity; establish the Forest Development Fund; provide for the implementation of the United Nations Framework Convention on Climate Change, Convention on International Trade in Endangered Species of Wild Flora and Fauna, the Convention on Wetlands of International Importance, especially as Water Fowl Habitat, the Convention on Biological Diversity, the Convention to Combat Desertification in those Countries experiencing Serious Drought and/or Desertification, particularly in Africa and any other relevant international agreement to which Zambia is a party; provide for matters connected with, or incidental to, the foregoing."

Relevance: The proposed area has flora that may be impacted during clearing for the establishment of pits, mine infrastructure and related buildings within the proposed project area, thus this Act is vital in ensuring protection of the flora species in the project area.

Compliance there of: The proponent will endeavor to act within the provisions of this law in the implementation of the project to protect forest resources within the project area

2.3.4 The Mines and Minerals Development Act (MMDA) No. 11 of 2015

The Zambian Regulation applicable to the development of a mine is the Mines and Minerals Development Act of 2015. This legislation regulates activities relating to mine and minerals operations including quarrying and provides for regulations for environmental protection during prospecting and mining activities and construction of the areas mined. It regulates for the conservation and protection of the environment and protection of human health and safety during prospecting, exploration and mining operations. Below are the applicable subsidiary SIs of the MMA that are relevant to African Fluorite Mining Limited project:

(a) Statutory Instrument No.29 of 1997 – Mines and Minerals (Environmental) Regulations – provides the framework for conducting and reviewing environmental impact assessment for the mining sector. Further to that, it provides regulations for auditing project implementation.

(b) Statutory Instrument No 102 of 1998 – Mines and Minerals Environmental Protection Fund Regulations – provides the mechanism of setting up and operating the Environmental Protection Fund. Thus, the mine will contribute to the fund so that if the mine owners left or abandoned the project, funds could be used to clean up the environment.

(c) Statutory Instrument No.7 of 2016- Mines and Mineral development (general) regulations- provides for the application of a license for exploration, mining, mineral processing to the Director of the Cadastre unit of the Ministry of Mines and Minerals Development.

(d) Mining Regulations (1971)

The mining activities shall be guided by the Mining Regulations which in essence are safety regulations. In accordance with the regulations' requirements African Fluorite Mining Limited "Holder" of the Prospecting Permit and Mining License shall appoint a Manager who shall be responsible for the day to day running of operations at the site including enforcing of the Mining Regulations and shall report incidences to relevant authorities such as the Mines Safety Department (MSD). The Holder shall provide the necessary resource to the Manager for the safe running of operations.

Relevance: The Act is relevant in ensuring the project activities are in line with the required standards as prescribed by the Zambian Law.

Compliance thereof: The proponent Mine Manager will subsequently appoint competent persons or supervisors in charge of various functions to ensure compliance to issues related to safe working procedures and use of proper Personal Protective Equipment by workers.

2.3.5 The Road Traffic Act No. 11 of 2011

An Act to make provision for the care, maintenance and construction of roads in Zambia, for the control of motor traffic, for the licensing of drivers and motor vehicles, for the compulsory third party insurance of motor vehicles, for the licensing and control of public service vehicles and public services, and for other miscellaneous provisions relating to roads and motor traffic.

Relevance: Haulage of construction material, deliveries of fuel especially for the operation phase will result in a considerable amount of additional traffic on the local roads and other local roads in the area. The Act is relevant to the project as the expected impacts of the project undertaking include increased traffic levels during delivery of materials during operation.

Compliance thereof: Signs and directions to control traffic movement to ensure a safe environment both in proximity to and within the site will be put in place. African Fluorite Mining Limited will make sure that there are designated routes to and from and within the site, built to national road standards.

2.3.6 The Explosives Act No. 10 of 1974

The Act defines explosives and the raw material used for manufacturing of explosives. It further provides guidelines for manufacturing, storage, transportation, distribution and use of all explosives and related products/accessories in Zambia. Safety issues are the most outstanding in this Act.

Relevance: The mining methods adopted **may** require procurement, transportation, storage and use of explosives.

Compliance thereof: The developer will conduct its operations in accordance with provisions of this Explosives Act and also ensure that the regulatory institution (Mines Safety Department) is fully aware.

2.3.7 Pneumoconiosis Act No. 13 of 1994

This Act (No. 13 of 1994) provides for the medical examination and standards of physical fitness to be required of persons exposed or likely to be exposed to the risk of pneumoconiosis. All mine employees that work in scheduled mine and scheduled

places will be required to undergo periodical medical examinations to ascertain their fitness to work in areas where humans are likely to be exposed to free silica in respirable dust, with particle sizes less than 5 microns, which become harmful when inhaled for over a long period of time. The Project work places and its immediate surroundings will be in the scheduled mine and scheduled places. Therefore, employees will be subjected to periodic examinations to ensure that they are physically fit to work.

Relevance: Due to the nature of the project workers will be exposed to the risk of pneumoconiosis.

Compliance thereof: the developer will endeavor to act within the provisions of this law in the implementation of the project by ensuring all workers are medically examined every year.

2.3.8 The Energy Regulations Act No.12 of 2019

An Act to provide for licensing of enterprises in the energy sector, continues the existence Energy Regulation Board and redefine its functions and powers; to provide for the licensing of undertakings for the production of energy or the production or handling of certain fuels; to repeal Energy Regulation act of 1995, and to provide for matters connected with or incidental to the foregoing.

The Energy Regulation Board (ERB) was established through the Energy Regulations Act No. 16 of 1995. The ERB is the sole licensing authority for operators in the energy sector and is responsible for close monitoring and supervision of such operators. It seeks to promote competition and ease of entry into the energy sector as well as safeguarding consumer interest.

Relevance: The storage of the fuel at the facility will be regulated by Energy Regulation Board (ERB).

Compliance thereof: The proponent will ensure that a license for the use of petroleum products is obtained.

2.4 International Agreements and Conventions

Zambia is a signatory to a number of international conventions. Conventions of significance to the proposed project are briefly described below.

1) Kyoto Protocol to the United Nations Framework Convention on Climate Change article 3

The Parties included shall, individually or jointly, ensure that their aggregate anthropogenic carbon dioxide equivalent emissions of the greenhouse gases do not exceed their assigned amounts, calculated pursuant to their quantified emission limitation and reduction commitments and in accordance with the provisions of this Article, with a view to reducing their overall emissions of such gases by at least 5 per cent below 1990 levels.

Relevance: The proposed project will involve vegetation clearing during construction phase and also operation of machinery.

Compliance: The proponent will ensure that no indiscriminate clearance of vegetation is done and that all machinery operating at the site are well serviced.

2) Basel Convention on the Control of Trans-boundary Movements of

Hazardous Wastes and their Disposal: article 10 sub section 1

The Parties shall co-operate with each other in order to improve and achieve environmentally sound management of hazardous wastes and other wastes.

Relevance: - hazardous wastes may be generated in the form of used oil or oil filters

Compliance: Maximum control measures shall be put in place to ensure that their transportation and disposal is done in accordance with local regulation

3) Convention on Migratory Species and the African -Eurasian Water Bird Agreement article IV:

Like other migratory species, water birds cross several international borders during their migration, facing a wide range of threats. Without international cooperation, conservation efforts of one country can be meaningless if these birds are not protected in another country.

Relevance: - The site is in an area of high bio diversity

Compliance: Maximum measures will be put in place to ensure that natural habitat of these birds is not completely disturbed

4) The United Nations Framework Convention on Climate Change (UNFCCC): Article 3.3.

Parties should take precautionary measures to anticipate, prevent or minimize the causes of climate change and mitigate its adverse effects. In order to ensure global benefits at the least possible cost, such policies and measures should take into account different socio-economic contexts, be comprehensive, cover all relevant sources, sinks and reservoirs of GHGs and adaptation and comprise all economic sectors.

Relevance: - land clearance will impact carbon sinks and emissions from construction equipment and will contribute to GHGs.

Compliance: Best environmental practices will be adhered to.

3.0 PROJECT DESCRIPTION

3.1 Nature

The proponent wishes to set up a mine that will be mining for the aforementioned ores using an open pit method on a 6.58Ha piece of land.

The mine will comprise the following facilities:

- Worker's quarters with kitchen 200m²;
- A store room 100 m²;
- A stockpiling and loading area 1500 m²
- Ablution block 10 m²
- Waste rock area 1500 m²

Mulonga Mining Cooperative Society Limited will undertake the following activities,

- Exploration and sampling activities
- Clearing vegetation for road and mining facilities
- Open pit mining;
- Stock piling and loading of mineral ore
- Transportation of mineral ore
- Management of waste rock
- Occasional blasting depending on underlying rock type
- Dust suppression

3.2 Project Objective

To undertake mining (Copper, Zinc, Cobalt, Gypsum and Felspar) activities that will take advantage of the mineral wealth of the area for economic gain.

3.3 Project Location

The proposed site is located in Mandale area under Chief Sikongo in Chirundu District of Southern Province. The site can be accessed using the Chirundu - Lusaka (T2) road and is about 5.45Km off the T2 road and about 24Km from Chirundu - Siavonga junction. Notable landmark is DPT towing company which is about 8.78 km South-East of the proposed mining site. The site lies on GPS coordinates 16°00'06.00S" and 28°35'12.00E".

Table 2: coordinates for project site

Corner points	Latitude Degrees (S)	Minutes	Seconds	Longitude Degrees	Minutes	Seconds
1	16	00	06	28	35	12.00
2	16	00	06	28	35	24.00
3	16	00	12	28	35	24.00
4	16	00	12	28	35	12.00

Mulonga Cooperative Artisanal Mining Project

proposed mine area

proposed mine area

Legend

Proposed mine site

Musaya straem

Great North Road

Google Earth

Image © 2024 Maxar Technologies
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Image © 2024 Airbus

2 km



3.4 Nature of the Project

The proponent applied for artisanal mineral rights for Citrine, Copper, Feldspar, Gold, Gypsum, Anhydrite, Lead, Manganese, Mica, Nickel and Zinc. Mulonga Cooperative wishes to set up an artisanal mine that will be mining for aforementioned ores using an open pit method on a 6.58Ha piece of land. The mine which is expected to produce about 100 tons of copper per day will comprise the following facilities:

- Worker's quarters with kitchen measuring 200m²;
- A store room measuring 100 m²;
- A stockpiling and loading area measuring 1500 m²
- Ablution block measuring 10 m²
- Waste rock area measuring 1500 m²

Mulonga Mining Cooperative Society Limited will undertake the following activities,

- Exploration and sampling activities
- Clearing vegetation for road and mining facilities
- Open pit mining;
- Stock pilling and loading of mineral ore
- Transportation of mineral ore
- Management of waste rock
- Occasional blasting depending on underlying rock type
- Dust suppression

3.5 Process and Technology

On commencement of mining operations, the Mineral ore will be removed from the ground using open cast mining (Open-pit – 90% of ore is mined using the open pit method). Ores near the surface are quarried after removal of the surface layers (overburden).

Currently, Copper will be the main target mineral. The Copper deposits will be mined from the open pit. The mining method that will be used for the open pit is the conventional selective open pit mining, where deposits will be delineated through benching and drilling of the deposits.

The process will include first waste removal to access the orebody, and then the ore will be moved accessed using excavators. Occasional drilling and blasting will be conducted in the case of hard rock Second the material will be loaded using hydraulic backhoe excavators and hauled using trucks and then taken to a mineral ore processing companies to be sold.

3.6 Raw Materials

This will include laterite, mineral ore, building sand, cement, water, crushed stones and steel reinforcement and steel structures, tiles.

3.7 Waste generation

Mostly general waste will be generated during both construction and operation stages. Hazardous waste may be generated as a result of cleaning of spillages or servicing of machinery on site. However, Machinery will not be serviced on-site and in the event

that it happens, used oils and oil filters will be generated which constitutes hazardous waste. The generated waste will be disposed of in accordance with local legislation.

For effluent, dewatered waste and run off mine as well as acid mine water will be channeled to a settling pond for settling to avoid the potential contamination of surface water with heavy metals.

3.8 Products

The goal is obtaining copper or the minerals of interest such Zinc, Cobalt, Feldspar and Gypsum. The mine is expected to produce around 100 tons of copper per day.

3.9 Schedule and life time of the project

The proponent wishes to proceed with project implementation as soon as ZEMA issues an approval and preferably by end of April, 2024. The project life span is projected at over 50 years.

3.10 Main project phases

The project will consist of three implementation phases namely; preparation, construction and operation. The phase will comprise the following activities;

3.8.1 Preparation phases

This phase will include the following activities:

- Obtaining relevant authorization from the Ministry of Mines and Minerals Development and ZEMA and other relevant stakeholders;
- Obtaining necessary expertise;
- Site clearing and preparation and;
- Procurement and transportation of the building material and equipment.

3.8.2 Construction phase

The phase will involve the following:

- Excavating and trenching works;
- Material backfilling;
- Compaction works;
- Concrete works encompassing building;
- Roofing.

3.8.3 Operation phase

At operation phase the main activity will be provision of the following services:

- Sampling and testing of mineral ores;
- Open pit mining using small tools and heavy machinery;
- Stock piling and loading of minerals ores
- Waste rock management
- Dust suppression
- Occasional drilling and blasting
- General waste management and
- Dewatering.

3.8.4 Decommission and closure phase

Activities to be undertaken during this phase will lead to the successful closure of the project while ensuring the biodiversity of the project area is maintained or enhanced. Activities will include:

- Notification of all relevant and affected stakeholders of the closure and decommissioning and;
- Removal of dilapidated infrastructure
- Pit stabilization
- Backfilling and re-vegetation of the areas
- Disinfecting and backfilling of septic tanks

4.0 PROJECT ALTERNATIVES

Project alternatives may comprise alternative Location, alternative process/design, water alternative, waste management alternatives and energy alternative. The EIA contributes to generation of a number of project alternatives.

4.1 Do nothing option

This option was not explored given the vast benefits to be accrued from implementation of this project.

4.2 Project Location Alternatives

The alternatives for site selection are limited due to the nature of the project which require mineral ore deposits. The alternatives included;

- Selecting an alternative site
- Implementation of the project at the proposed site

Another site

The proponent realized that looking for another site will require finding another site with similar mineral deposits which are not being mined

Proposed site

The proposed site which is already explored and has been known to have mineral deposits which have not been mined.

Chosen Option

The **second option** was **chosen** as it was deemed economically feasible.

4.3 Mining Methods

Two options were explored. That is, open pit and underground Mining. However, being artisanal mining, open pit was the chosen method due to its lower costs and risks. Also, the minerals of interests were found to be near the surface.

4.4 General waste Management

Solid waste to be generated at the mine will mainly be packaging plastic material, left over foods stuffs and paper waste.

Waste management alternative considered included **onsite incineration**, transporting of waste to a ZEMA approved disposal site and disposal in an onsite pit.

Chosen option

Transportation to a ZEMA approved disposal site was the chosen option for non-biodegradable waste. Bio degradable waste will be composted at the site.

4.5 Energy Source

Three alternatives were considered for the project, these included using a diesel-powered generator, connecting to the national grid or use of solar energy. Solar

energy and diesel-powered generator energy were the chosen options for the time being.

4.6 Sewage management

Two options of sewerage treatment were considered for the project. This included construction of a septic tank and use of mobile toilets.

Chosen Alternative

Septic tank construction was the chosen option

5.0 DESCRIPTION OF THE BASELINE ENVIRONMENT

The proposed project area is in Chirundu District of Southern Province.

5.1 Climate

The climate at the site is typical of the Southern province region with dry, wet and hot seasons during the different times in a year.

Temperature

The area has the Valley's cool "winter" months of May to August. Temperatures fall to about 10° Centigrade in June and July. Generally, night-time temperatures during this time are 15 °C and rising to 25 °C during the day. Days and nights are clear and the surface wind is generally light and predictable. From October it is hot and dry. Daytime temperatures build up to 40 °C dropping to 30 °C at night in some high years. In November the chance of rain improves and the trees push out their new buds to lie in wait for the first rains. Air temperatures are usually above 20° C, but have a distinct seasonal variation. Maximum temperatures are in October and November when mean monthly maximum is 33-38 ° C and absolute maximum of 41 ° C and gradually decrease during the rains to a minimum in June/July mean monthly maximum of 13 °C and an absolute minimum of 3 °C. Daily hours of sunshine vary from a mid rainy season low of 6.4 hours to a winter high of 10.1 hours.

Annual Rainfall

Mean seasonal rainfall varies from 680 mm to 950mm, but can be as low as 356 mm and as high as 1299 mm in some years. The unreliability of rainfall in this area is illustrated by three forms of drought; from April to October; frequent dry spells in the rainy season and conventional droughts of low seasonal rainfall as indicated above.

Soils

The soils of area are ferrallitic in nature and are derived from various parent rocks, including granite, gneiss, sandstone and schist. These are old soils and are slightly leached due to high wind and medium rainfall with lower base saturation of pH between five and six, having a red or reddish-yellow colour or gray colour. The clay content is generally higher: soil textures change from clays to sandy clays or sandy clay loams, showing gradually increasing clay content from the coarse-grained surface soil to the sub-soils. Although they are clay-dominated soils with low permeability, they are generally well drained.

Flora

The project area falls in predominantly *Acacia albida*-dominated woodland that has however been thinned out due to previous human and livestock impacts and is better described as mixed *Acacia albida* / *Combretum berbe* / *Kigelia africana* / *Lonocarpus capassa* woodland with isolated stands of *acacia albida* and thick herbaceous understorey of *senkenene* (*Paspalum paspaloides*). The surrounding area consists of degenerated *colophospemum* / *Kirkia* / *Terminalia* woodland and grassland on the bottom close to shore.



Figure 2: vegetation at project site

Fauna

Common in the regions are velvet monkeys (*Chlorocebus pygerythrus*) and baboons (*pipío*) which are found in the inland areas of Chirundu. Velvet monkeys have been spotted around the project area too. Lake Kariba is also rich in fishes as well as big game like crocodiles. The other fauna are birds which are common in the area. There seems to be no quantitative data on these species. Other fauna observed in the project area included domesticated animals such as cattle. (*Bos taurus*)



Figure 3: fauna in the nearest settlement

Air Quality and Noise

The mine site falls in a greenfield with no habitation and industries. The air quality in the project area is generally good. The only source of air pollution is vehicular movement from the road about x km away and charcoal burning activities..

Water Resources

Chirundu District is serviced by the Zambezi river and on a small scale the Kafue River. These are the two main drainages in the District. The mine area has a non-perennial stream called Musaya which is about a kilometer from the mine site. There are no other sources of water in the surrounding areas and mostly depend on boreholes and shallow wells from domestic water.

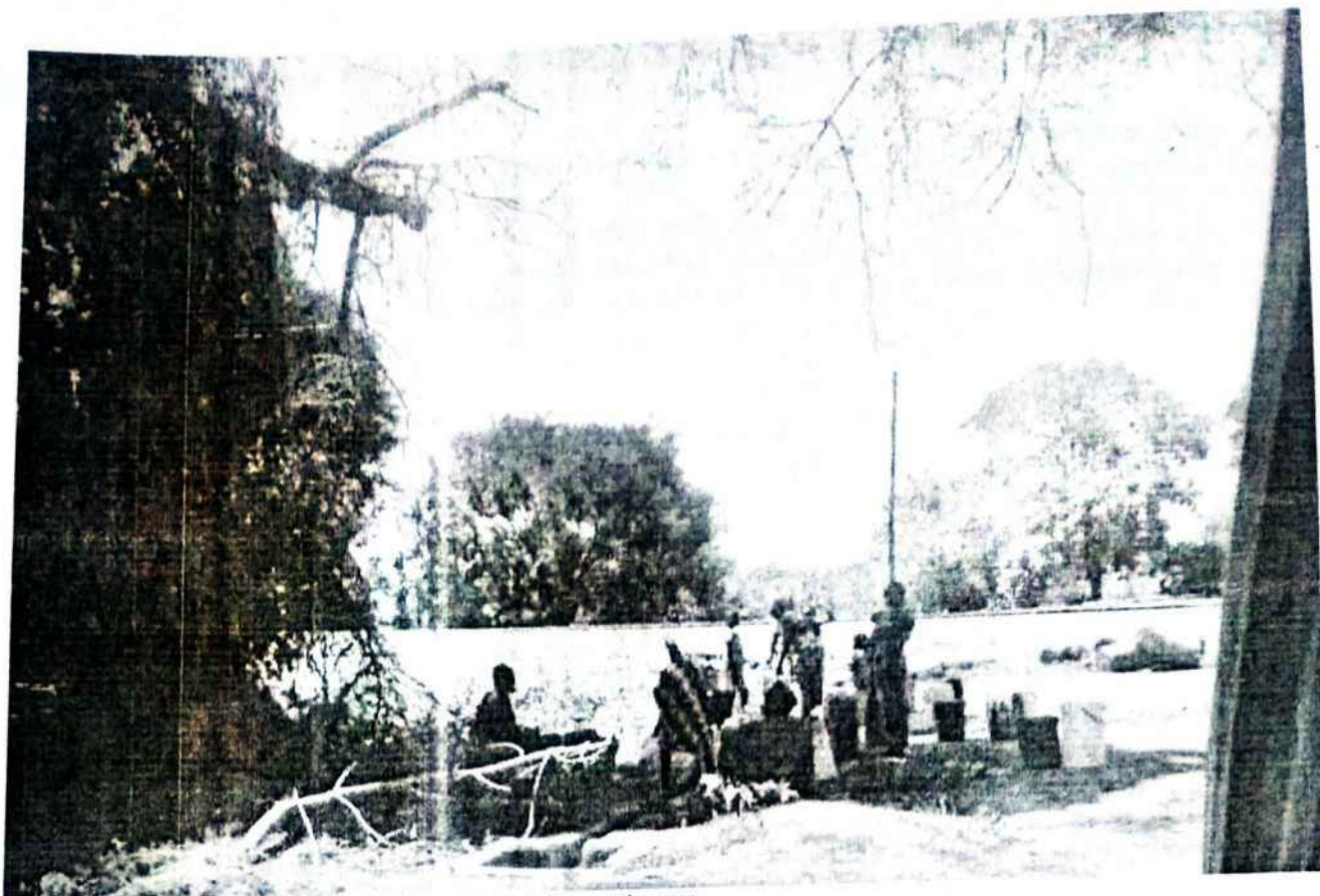


Figure 4: communal water point in the nearest surrounding settlement

Water Quality

The only water surface point in the area is the non-perennial Musaya stream which was almost dry at the time of the field study. The stream is mainly used as a source of water for animals as well as gardens. The water quality is generally good as there are no effluent discharging activities in the area. No water samples were collected as there are no water sources currently nearest the mine site which would give representative data of the baseline.

Administration

Like the rest of the country, there are three administrative systems in the District and these are: Central Government, Local Government (Authority) and Traditional Systems. Central Government Administration is composed of all government departments and is headed by the District Commissioner (DC) appointed by the Republican President. The DC supervises and coordinates government activities in the district. He also harmonizes the activities of the Local Authority with those of Central Government in the Districts. Heads of Departments in respective government departments in the Districts report directly to their provincial heads on technical matters and to the DC on administrative matters. Local Government Administration system is based on the political delineation of the District.

The constituency in the affected area has a duly elected member of parliament who represents the residents in the National Assembly. The Member of Parliament has a mandate covering a period of five years. Councillors are also elected to represent various wards. While the Councillors provide political Direction of the District, the executive arm of the District is headed by a Council Chairperson who is assisted by directors covering various functions of the council. The Council Chairperson is directly responsible for the execution of Council development strategies and also provides technical and professional

guidance to the Council on matters relating to District development. The Council however, endorses what is to be implemented based on resolutions from its full Council meetings.

The traditional leadership in the area is headed by Chief Sikongo. Through the traditional lineage and ascendancy guidelines, the chief is flanked by his traditional councils. These are appointed by the Chief and can be dissolved by the Chief as well. These councils head a grouping of villages, who in turn are headed by Village headmen. These headmen are also subordinated by village headman. The Village headmen and Council leaders are close advisers to the Chief.

Agro-Ecological Zone

Chirundu is in Agro-ecological Zone I, constitutes about 12 percent of the country and covers Luangwa and Zambezi river valleys. It has low altitude and low erratic rainfall with a short growing season. Although it is hot and humid with poor soils in some places, it is suited for growing drought resistant crops. Goat rearing and fishing potential are high as is cattle rearing and cashew nut production in west of Zambezi River in Western Province.

Archaeological, Cultural and Historical resources.

The UN Foundation (2004) argue that Heritage Sites carry out numerous functions that are beneficial and essential to human welfare. The nearest cultural and heritage site is the Chirundu Fossil Forest

Livelihood

The local people mainly derive their livelihood from agriculture. Pastoral farming and Tourism are the predominant activities in the District. Maize, sorghum, are some of the most grown crops. Formal employment opportunities are mostly centralized at Chirundu border. However, In Mandale village and surrounding areas, the main livelihood is Charcoal burning and selling and pastoral farming. However, the locals have also started engaging in illegal mining activities in the project and surround areas.

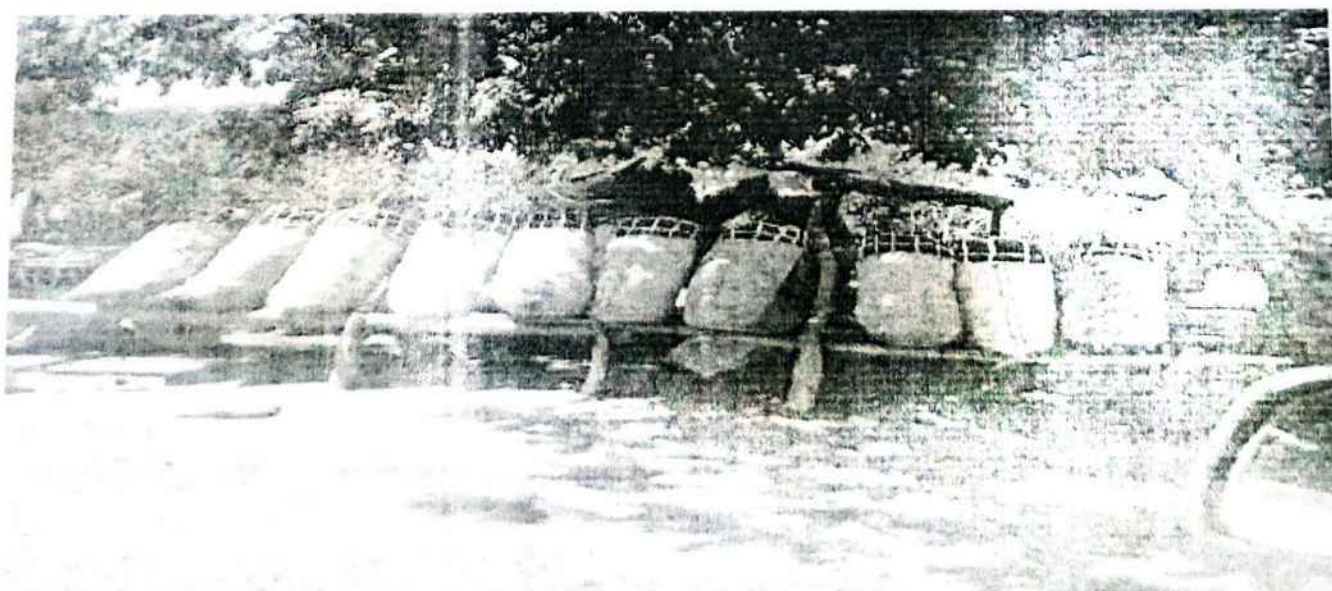


Figure 5: Charcoal selling in the nearest areas

Education and Healthcare

The Mine site is in Mandale village of Sikongo Chiefdom. The nearest school to the area is Katwezele Community School which is over 5Km from the mine site. There are no health care facilities in the project area. The Nearest Health clinic is in Siagweemu village.

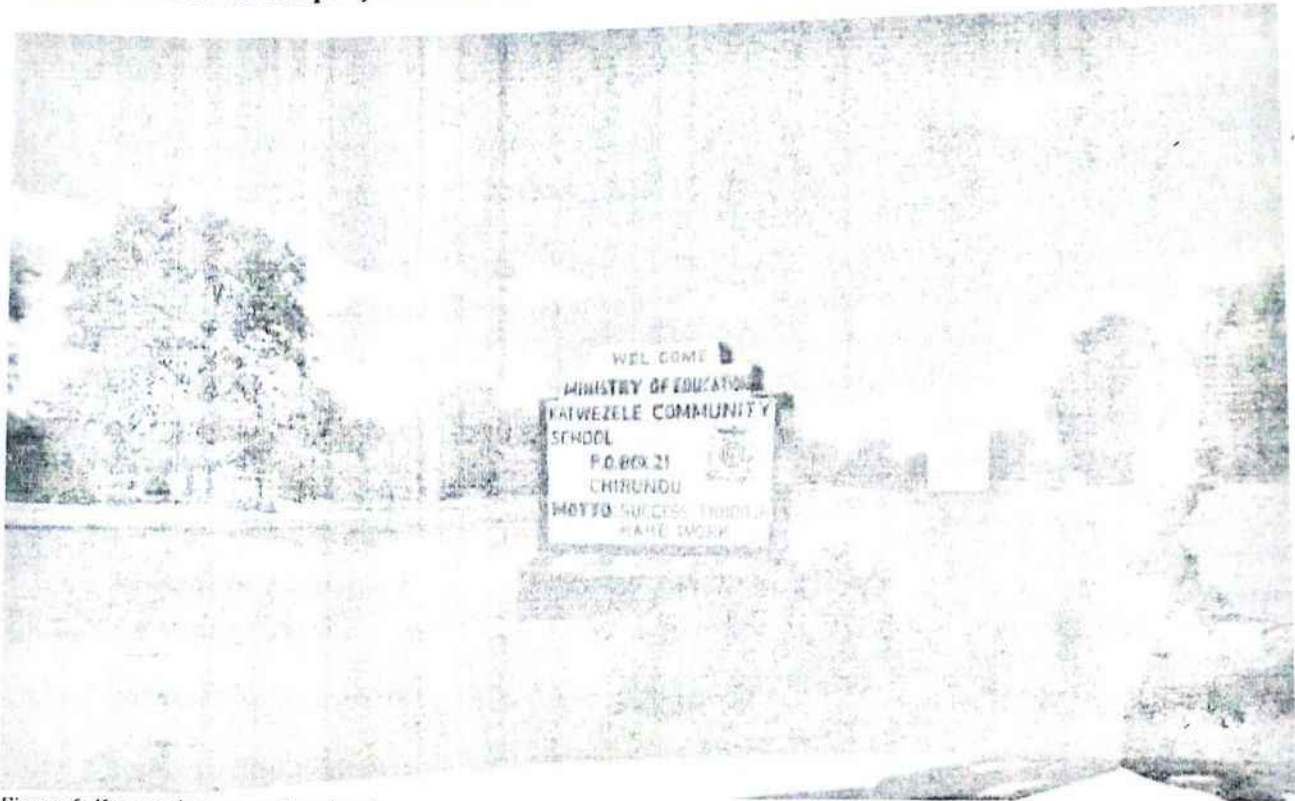


Figure 6: Katwezele community school

Illegal Mining Activities in the area

The area has some illegal mining activities taking place. Illegal miners use simple mining tools of hammers and picks to break rocks and stock pile.



Figure 7: Results of illegal mining in the area

6.0 ENVIRONMENTAL AND SOCIAL IMPACTS

The environmental and social impacts associated with the proposed development of a mine are likely to trigger both negative and positive impacts. This requires that the negative impacts be managed and the positive impacts be enhanced to minimise the potential impacts on the environment. The evaluation of impacts was based on the methodology described in the subsequent section below.

6.1 The Nature and Evaluation of Impacts

Studies undertaken considered direct, indirect, cumulative, and residual environmental impacts associated with the construction and operations of a mine. These were assessed in terms of the following criteria: description of what causes the effect, what will be affected, and how it will be affected. The extent, wherein it is indicated whether the impact will be local (limited to the immediate area or site of development), regional, national or international. The following factors were considered in classifying each potential impact generated by the Project.

- **Frequency:** Occurrence of activity producing the impact, e.g. continuous, intermittent or a single event/less than once per year;
- **Likelihood:** Probability of impact occurrence (e.g., 100%, 50%, 0%);
- **Extent:** Spatial extent of the impact (e.g. within 2km of site boundary, outside the Project site but within 20km, within 200km, within Zambia, outside Zambia.
- **Duration:** Extent in time of the impact. Short term impact (less than the life of the project), medium term impacts (equal to the lifetime of the Project) and long-term impacts (greater than the lifetime of the Project);
- **Magnitude:** Impact magnitude defined in relation to the limit criterion specified by ZEMA or international standards where available.
- **Type of impact:** Positive or negative effect; direct or indirect action.
- **Potential significance:** A combination of all the factors described in the preceding bullet points is used to determine the type and significance of potential impact prior to mitigation. This is defined as low, medium or high.

6.2 Evaluation of Potential Impacts

An attempt is made in this section to evaluate the overall significance of each identified impact by assigning values to a set of criteria as follows:

Table 3: Significance ranking matrix

CONSEQUENCE (Magnitude Geographic Extent Duration of impact)															
Likelihood of impact	1	2	3	4	5	6	7	8	9	1	1	12	13	14	15
	2	4	6	8	10	11	14	16	18	20	22	24	26	28	30
	3	6	9	12	15	18	21	24	27	30	33	36	39	42	45
	4	8	12	16	20	24	28	32	36	40	44	48	52	56	60
	5	10	15	20	25	30	35	40	45	50	55	60	65	70	75
	6	12	18	24	30	36	42	48	54	60	66	72	78	84	90
	7	14	21	28	35	42	49	56	63	70	77	84	91	98	105
	8	16	24	32	40	48	56	64	72	80	88	96	104	112	120
Frequency	9	18	27	36	45	54	63	72	81	90	99	108	117	126	135

Table 4: mitigation rating

Significance Rating	Value	Negative Impact Recommendation	Positive Impact Recommendation
Very High	126-150	Propose measures mitigation	Improve current management
High	101-125	Propose Measures mitigation	Improve current management
Medium-High	76-100	Propose Measures mitigation	Improve current management
Low-Medium	51-75	Maintain current Management	Propose Measures mitigation
Low	26-50	Maintain current Management	Propose Measures mitigation
Very low	1-25	Maintain current Management	Propose Measures mitigation

- **Type of Impact** (Direct / Indirect)
- **Duration** (Intermittent / Temporary / Short Term / Medium Term (construction phase) / Long Term / Permanent)
- **Intensity / Severity** (Negligible / Low / Moderate / High)
- **Spatial Extent** (Site / Local (area surrounding the site) / Regional)
- **Likelihood** (Uncertain / Improbable / Probable / Highly Probable / Certain)
- **Sensitivity** [of the receptor; degree of change effected on natural processes or people's livelihoods] (Low / Moderate / High)
- **Overall Significance** (Insignificant / Low / Moderate / High)

6.3 Anticipated Project Impacts

The positive and adverse project impacts assessment was based on the following impact criteria during pre-construction, construction, operation phases and decommissioning phase as follows:

- **Direction of Impact:** Type of effect the activity would have on the affected environment,
- **Positive or negative**
- **Scale:** Physical scale / area over which the impact will be felt: Local, Regional, National or International
- **Duration:** The length of time the impact is likely to occur: Short, Medium or Long Term
- **Reversibility:** Reversible; where it is avoidable or has potential to be restored; Irreversible; where it is unavoidable or has no potential to be restored.
- **Pathway:** Indirect; where it is no direct result of the Project. Direct; where it is a direct result of the Project and associated activities; Cumulative; where it results from incremental changes caused by other past, present or reasonably foreseeable actions together with the Project

Positive Environmental and Social-Economic Impacts

The implementation of the project activities described in this report will lead to numerous positive social impacts during the various phases of the project. These will include;

- Employment creation during all phases of the project;
- Contributing to the diversification of the economy of Zambia;
- Increased revenue for the central government through corporate taxes and license fees and;
- Improvement of the living standards of the local people through multiplier effects.
- Increased presence of people will reduce cases of poaching

Negative Environmental and Social-Economic Impacts

6.3.1 Surface and underground water

Surface water hydrology can be affected during the construction and operation of the proposed development and these impacts may continue following cessation of activities. Construction activities can result in compaction of soils, an increase in impermeable (or slowly permeable) surfaces and the installation of drainage. The subsequent increase in surface runoff may, in turn, increase the risk of silt. Further contamination can result from oil spillages and leakages and acid mine drainages

Mitigation measures

- All spills will be cleaned and stored in closed containers and treated by licenced environmental companies.
- The wastewater generated from the toilets and kitchen, will be directed to the septic tank.
- Bins will be provided to prevent indiscriminate disposal of waste
- Construction of a settling pond to capture acid mine drainage, silt and any other waste.

6.3.2 Air pollution

Potential to affect local air quality somewhat as a result of dust from mines, emissions from as NO_x and NO₂ from blasting activities. Also, air quality will be affected by vehicular emissions and poor waste management.

Mitigation measures

- During construction works the site shall be constantly sprayed with water in order to suppress the dust.
- Workers will be provided with protective clothing such as masks.
- Waste will be managed adequately to prevent foul smells
- No open-air burning will be allowed within the project site
- Sewer pipes and septic tank to be adequately maintained and monitored to prevent leakages
- Operation times will be limited to day time.

6.3.3 Impact on Biodiversity and Vegetation

- Loss of Vegetation: Preparation/construction activities may lead to loss of vegetation due to clearance of the area for the project development.
- Loss of habitat for fauna due to vegetation clearance, soil compaction and human settlement
- Disturbance of the food chain due to unnatural interference from human activities
- Loss of fauna due to noise and human presence

Mitigation measures

- No indiscriminate vegetation clearance will be permitted. Vegetation clearance will be limited to area of interest
- No poaching of any form of hunting will be done

- Noise levels will be maintained to less than 85db and no form of loud music or loud machinery will be operated during operations phase
- Number of visitors at the site will be kept at low levels at all times to reduce human-animal conflict

6.3.4 Solid Waste Generation

Solid and Liquid waste will be generated during construction and operation phases of the project. This will include cleared vegetation, topsoil, rubble, domestic and sanitary waste

Mitigation measures

- Rubble produced during construction stage will be re-used in landscaping of the area and backfilling of trenches.
- Clearance of vegetation will be limited to areas of interest and no indiscriminate clearance will be done.
- Septic tanks will be emptied on time when full
- Bins will be provided for domestic waste which will be eventually be transported to designated sites.
- Top soil will be re-used for revegetation and flower beds

6.3.5 Liquid waste generation

This will include sanitary waste, effluent from the fish pond and kitchen waste that will have the potential of polluting land and water sources. These can even impact air quality if left unattended to through foul odor emissions.

Mitigation measures

- All sanitary was and kitchen effluent will be directed to the septic tanks which will be emptied on time when full
- All run-off dewatered water from the mine will be directed to a settling pond

6.3.6 Fire and explosions

Negligence, poor management and poor wiring can lead to fire outbreaks.

Mitigation measures

- All electrical connections shall be done by competent people who will be doing routine checks
- Fire extinguishers will be placed at strategic points.
- All workers will also be trained in firefighting skills.
- Smoking zones will be properly be labeled especially for the petroleum storage area.
- All explosives will be managed by designated personnel only and storage area fully secured with warning signs. Explosives will be stored and transported separately from detonators.

6.3.7 Community Health and Safety

The proposed project might affect the surrounding communities due to the interactions between the tourists and the locals leading to communicable diseases such as STIs, Covid and others. Also, poor sanitation may lead to poor

health.

Mitigation measures

- Education on health to be conducted thoroughly
- Provision of condoms and where necessary masks in strategic area to be done
- Promotion of good sanitation,

6.3.8 Occupational health and safety risks

The proposed project involves the workers interaction with various machinery and tools and exposure to risky tasks and areas. If not well managed the workers might incur injuries, burns, as well as death.

Mitigation measures

- Fire extinguishers and first aid boxes will be placed at strategic points.
- Workers shall be given awareness talks on safety
- The Developer shall provide and maintain necessary personal protective equipment
- Warning signs will be displayed around the site.
- Working hours will conform to those prescribed by law to prevent stressing the workers
- No feeding of animals will be done
- Only qualified personnel will be allowed to conduct specialized jobs
- Proper mining methods will be used to avoid mine failure
- Waste rock sites will be managed by qualified personnel to ensure proper stacking and inclinations.

6.3.9 Hazardous waste

There is a likelihood of having spillages during construction and operation activities through spills, leakages and servicing.

Mitigation measures

- The servicing of the equipment will be done at nearby workshops to prevent any probable leakages onsite
- On site repairs will be supervised and done in an environmentally friendly manner
- During construction there will be emergency used oil collecting vessel in order to collect any leakages. The oil after collection will be collected by Licensed companies.
- The contaminated solid waste or any materials shall be cleaned using degreasers before being disposed of according to the provisions of the law.

6.3.10 Impact on Traffic and road Safety

During all the stages of the project there is a likelihood that disturbance to the normal flow of traffic in the access road might occur.

Mitigation measures

- Traffic signs and speed limits to be put up along the mine area
- Only Licensed driver to be employed and allowed to drive

- Only designated routes to be used by the mine drivers
- Sobriety tests to be conducted on the drivers

6.3.11 Soils

Soil will be impacted throughout the project phases through: compaction, vegetation clearance, oil leakages and top soil removal. This might lead to soil erosion, loss of biodiversity and water contamination.

Mitigation measures

- During construction: The servicing of machinery and equipment will take place away from the site
- Construction works will take place during the dry season in order to easily manage the loose soils
- During operation: Drainages will incorporate silt traps for easy management of site storm water.
- No open-air burning will be conducted on site
- Licensed experts will be engaged to clean and treat contaminated soils
- Construction machinery and operational cars will only use designated paths to avoid unnecessary soil compaction.
- Top soils removed during construction works will be stored properly and reused during landscaping

6.3.12 Human-Animal conflict

As the project is located in an area that is sometimes visited by wildlife, it is inevitable that humans will interact with wild animals. These interactions, may bring about conflict resulting in catastrophic results such as loss of fauna, loss of human life and injuries

Mitigation

- All guidelines provided by the DNPW will be adhered to
- There will be no feeding of wild animals at or around the project site
- All speed limits will be adhered to and signage put up
- All waste will be secured and professionally managed
- No open-air burning will be conducted at the sight
- No poaching activities will be conducted and any such activities will be reported to relevant authorities

Table 5: evaluation of impacts

Impact	Sub impact/Impact Description	Impact Evaluation							
		Frequency	Likelihood	Extent	Duration	Magnitude	Effect	Sensitivity	Significance
Biophysical Environment									
Water Quality Impacts	Siltation	Infrequent	likely	Local	Short term	Medium	Negative	Low	Low
	Water contamination from Spent oils, oil Leakages, effluent from fish pond and accidental Spillages	Infrequent	likely	Local	Medium Term	Low	Negative	Low	Low
	Contamination of surface water by Acid Mine Drainage	Infrequent	Likely	Local	Medium term	Low	Negative	Low	Low
Erosion of Cleared areas	Exposed soil after excavation is vulnerable to erosions and runoffs by rains. Construction and operation activities could result in local inundation of the site and erosion problems in periods of rainfall.	Infrequent	Likely	Local	medium	medium	Negative	medium	Medium

Impact	Sub impact/Impact Description	Impact Evaluation							
		Frequency	Likelihood	Extent	Duration	Magnitude	Effect	Sensitivity	Significance
Biophysical Environment									
Increased vibration and Noise	Construction and mining activities generates high levels of noise and vibrations. These disturbances are mainly generated by heavy	Frequent	Likely	Local	Long term	Medium	Negative	High	medium

Impact	Sub impact/Impact Description	Impact Evaluation							
		Frequency	Likelihood	Extent	Duration	Magnitude	Effect	Sensitivity	Significance
Biophysical Environment									
	Machinery, blasting and drilling								
Pollution of Soil	Oil and fuel spills/leaks from machines may lead to soil pollution	Infrequent	unlikely	Site	Medium	Low	Negative	Low	Low
Loss of Biodiversity	Clearing of vegetation on site will lead to loss of habitat for flora and fauna.	frequent	Likely	Local	Long	Low	Negative	High	Medium
Generation of non-Hazardous Waste	Non-Hazardous/solid waste will mainly come from discarded Packaging materials such as carton boxes and plastic packs. Although the impact from this waste is expected to be minimal, poor disposal methods can lead to environmental problems due to their	Frequent	Likely	Site	Long	Low	Negative	Medium	Low

Impact	Sub impact/Impact Description	Impact Evaluation								
		Frequency	Likelihood	Extent	Duration	Magnitude	Effect	Sensitivity	Significance	
Biophysical Environment										
	non- biodegradable nature.									
Disposal of non-process general wastes	Inappropriate storage and disposal of general wastes, particularly leftover food stuff and food waste	Infrequent	Likely	Site	Short Term	Low	Negative	Medium	High	
impact on occupational health and safety and workers rights	Accidents to personnel as a result of operation of machinery and construction tools	Infrequent	Likely	Site	Long	Medium	Negative	Medium	Medium	
	Attacks from wild Animals	Infrequent	unlikely	Site	Short Term	Low	Negative	low	Low	
Impact on community health and safety	The nuisance of dust and the movement of construction machinery around the site may affect the safety and health of the community.	frequent	Likely	Local	Long term	Medium	Negative	Medium	Medium	
	There is a risk of accidents occurring involving members of	Infrequent	UnLikely	regional	Long term	Low	Negative	Low	Low	

Impact	Sub impact/Impact Description	Impact Evaluation							
		Frequency	Likelihood	Extent	Duration	Magnitude	Effect	Sensitivity	Significance
Biophysical Environment									
	the public who may gain unauthorized access or inadvertently venture on to the construction site.								
Increased employment opportunities	It is anticipated that the project will provide direct employment Opportunities during both site preparation and construction phases.	Frequent	Likely	Local/ Regional	Short Term	High	Positive	High	High
Increased public revenue/ taxes	The implementation of the project will increase revenue and taxes for both the central and local authorities through taxes, permits and fees	Frequent	Likely	Local/ Regional	Long Term	High	Positive	Medium	Medium

Impact Evaluation									
Impact	Sub impact/Impact Description	Frequency	Likelihood	Extent	Duration	Magnitude	Effect	Sensitivity	Significance
Biophysical Environment									
In-Migration	The project may potentially result in in-migration near the project area for those seeking employment	frequent	Likely	Regional	Term term	Low	Negative	Low	Low
	Temporary marriages, casual Sex relationships and more chances of transmission of Sexually Transmitted diseases as well as transmission of covid-19.	Frequent	Likely	Local	Long Term	Low	Negative	Medium	Medium
Boost of the local economy in the area	The presence of mineworkers and other related staff will boost the market for some products in the area	Frequent	Likely	Regional	Short term	Low	Positive	Medium	Medium

Table 11: shows the evaluation of impact during the operations phase

Table 11: Shows the evaluation of impact during the operations phase									
Impact	Sub impact/Impact Description	Impact Evaluation							
		Frequency	Likelihood	Extent	Duration	Magnitude	Effect	Sensitivity	Significance
Biophysical Environment									
Water Quality Impacts	Sanitary wastewater from facility.	frequent	unlikely	Local	Short term	Low	Negative	Low	Low
	Acid Mine Drainage	Infrequent	Likely	Local	Long	Low	Negative	Medium	Medium
Stress on water Resource	Abstraction of water from borehole	Frequent	Likely	Local	Long term	Low	Negative	Low	Low
Pollution of Soil	Oil and fuel spills/leaks from machinery	Infrequent	Unlikely	Site	Short Term	Low	Negative	Low	Low
	Soil compaction leading to loss of biodiversity	frequent	Likely	Site	Long term	Medium	Negative	Medium	Medium
Generation of non-Hazardous Waste	Waste water and solid Waste disposal	Infrequent	Unlikely	Site	Short Term	Low	Negative	Low	Low
	Improperly managed general waste will lead to increased human-life conflict	infrequent	Likely	Local	Long term	Medium	Negative	Medium	Medium

Impact	Sub impact/Impact Description	Impact Evaluation							
		Frequency	Likelihood	Extent	Duration	Magnitude	Effect	Sensitivity	Significance
Biophysical Environment									
Employment/Liability	Both direct and indirect employment will result from the	Frequent	Likely	Regional	Long	Medium	Positive	High	High

Impact	Sub impact/Impact Description	Impact Evaluation							
		Frequency	Likelihood	Extent	Duration	Magnitude	Effect	Sensitivity	Significance
Biophysical Environment									
	project for both skilled and unskilled workers								
Revenue For the Governm ent	The project will provide increased revenue to through various corporate Taxes and Licences.	Frequent	Likely	Regional	Medium Term	Medium	Positive	High	High
Improve d Local Capacity / Economy	The local capacity/economy will be improved by the project in that more investment will be attracted to the area	Frequent	Likely	Local	Long Term	Medium	Positive	High	High
Loss of access to	Loss of animal corridors	Frequent	Likely	Local	Long Term	High	Negative	medium	Medium

natural resources	Animal attacks	Infrequent	Likely	Local	Long term	Medium	Negative	Medium	Medium
Increased Traffic	Potential loss of fauna	frequent	Likely	Local	Long Term	Medium	Negative	Medium	Medium
	Accidents and dust emissions	infrequent	Likely	Local	Long term	Medium	Negative	Medium	Medium
Land degradation	Mining activities will result in creation of pits	frequent	Likely	Local	Long	High	Negative	Medium	Medium

Impact	Sub impact/Impact Description	Impact Evaluation							
		Frequency	Likelihood	Extent	Duration	Magnitude	Effect	Sensitivity	Significance
Biophysical Environment									
Fire	Loss of fauna and flora	Infrequent	Unlikely	Local	Medium term	High	Negative	Medium	Medium
Human-Animal conflict	Human - Animal attacks	Infrequent	Likely	Local	Medium Term	High	Negative	High	High

7.0 ENVIRONMENT AND SOCIAL MANAGEMENT PLAN (ESMP)

This section discusses the proposed enhancement and mitigation measures for each respective positive and negative Impact identified in the previous section. The proposed mitigations measures designed to minimize or eliminate for impacts triggered by the project.

The ESMP will include all the measures that are required for environmental protection, which will include the mitigation measures and the monitoring plan. The aim of an environmental management plan is to avoid, minimize, or ameliorate effects or impacts resulting from project implementation and where possible, enhance beneficial effects.

Positive Social-Economic Impacts

The development of this site and construction activities described in this report will lead to numerous positive social impacts during the various phases of the project;

- Increase in employment for the local people,
- Contributing to the diversification of the economy of Zambia,
- Increased revenue for the central government through corporate taxes and license fees,
- The project will improve the livelihood standards of the local people through multiplier effects.
- Opportunities for trade

7.1 Impact Mitigation Plan

The impact mitigation plan allocates the responsibilities for implementation of the proposed mitigation measures to the various stakeholders and indicates at what stage in the project they should be performed. The key components of the proposed impact mitigation are as following:

7.1.1 Impact on Water Resource

- Ensuring that the septic is located away (over 50m) from water sources and are continuously monitored.
- Ensuring that all vehicles are properly serviced to prevent leakages and all fuels are stored in secure places which are bunded.
- Installation of silt traps along drainage systems
- Ensuring that effluent is not released directly into water sources
- Continuously monitoring water quality parameters
- Channeling Acid Mine drainage to a settling pond

7.1.2 Impact on Biodiversity and Vegetation

- Restrict clearing of vegetation only to areas where mine, road and other structures will be established
- In general: strengthen the awareness of the workforce for the

environment (protected areas, plants and wildlife): to avoid logging, introduction of invasive species, hunting, etc.

- No open air burning on and around the project site
- Ensuring that only designated routes are used

7.1.3 Impact on Land and Soil

- Top soils removed during construction will be reused for revegetation. No mitigation is possible.
- Solid waste generated will be properly stored and transported to ZEMA approved disposal sites.
- Accidental fuel/oil spillage will be cleaned according to local regulations and storage of oils/fuels will be on concrete lined floors which are bunded.
- Maintenance of vegetation will be done to preserve ecosystems and prevent soil erosion
- Overburden will be well maintained and reused for backfilling at end of mine

7.1.4 Impact on Air Quality

- Watering of the area and surroundings during the construction and operation stages will be undertaken regularly
- Site workers will be issued with personal protective attire.
- All the sand or soil heaps will be removed as soon as possible to avoid nuisance dust arising from prevailing winds.
- All earth materials will be transported moist and/or covered with appropriate material to suppress dust.
- No open-air burning will be done on or around the project site

7.1.5 Waste Generation and Management

- A waste management system will be put in place for segregation, storage and disposal
- No indiscriminate disposal of waste on site, along the roads or in the surrounding areas shall be allowed.
- In general waste will be reduced, re-used, recycled and the disposal should be controlled.
- All equipment servicing will be done offsite. In case of onsite repairs, all generated waste will be handled as stipulated by law.

7.1.6 Noise

- All mobile vehicles and equipment will properly be serviced
- All land preparation activities will take place during the day and any work during night-time will be communicated to the state authorities and local community
- Well-maintained construction and transportation equipment will be used.

- Workers exposed to excessive noise will be equipped with PPE(e.g., ear protectors) and the exposition time has to be limited

7.1.7 Community Health and Safety

- Employing of local residents to minimise the spread of communicable diseases.
- The Construction Supervisor in collaboration with Ministry of Health shall take measures to educate and sensitize the labour force on the risks of communicable diseases (malaria, TB, STDs, including HIV/AIDS etc.).
- Masks and social distancing will be practiced and sanitizing materials will be provided in compliance with covid-19 requirements.
- All pit will be dewatered and water channeled to settling ponds. Settling ponds will be treated to avoid proliferation of mosquitoes.
- All acid mine drainage will be channeled to a settling pond to avoid discharging heavy metals into surface water bodies
- All explosives will be secured to avoid theft

7.1.8 Traffic and congestion

- Warning and traffic signs will be put up throughout the project cycle
- During operation, transport officer will assist in directing traffic in and out during peak periods
- Proper signage will be put up and speed limits will be adhered to
- Construction and operation works to be limited to day time

7.1.9 Fire outbreaks

- A no smoking policy outside designated smoking area will be adhered to
- All naked wires will be worked on
- Fire extinguishers will be made available at the project site and will be serviced
- No open-air burning will be done at the project site
- All petroleum will be stored in properly secured store rooms with proper signage

7.1.10 Human- Animal conflict

- All waste will be properly stored in waste bins and transported to designated waste disposal sites
- No poaching policy will be implemented
- Animal corridors will not be disturbed
- Speed limits will be adhered to
- Noise levels will be kept at a minimum

Table 6: ESMP

Aspect	Impact	Mitigation Measure	Frequency of Monitoring	Time Frame	Performance Indicator	Monitoring and Reporting	Cost (ZMK)
Loss of Flora	Loss of vegetation and biodiversity	Revegetation to be done	Monthly	On-going	- No signs of indiscriminate tree cutting. - Regeneration of trimmed/cleared vegetation	Manager / Contractor	1,000
Water Quality	Contamination of surface and groundwater	- Septic tank to be constructed away from borehole - Fuel storage to be done on designated areas which are concrete lined and banded - All spills to be cleaned up by spill kits - Construction of a concrete base with a bund wall around	Bi annually	On-going	Baseline water quality maintained	Contractor/Manager	2500
			Weekly	Short term	No signs of fuel and oil discharge in water and on soils	Manager/contractor	2,000

Aspect	Impact	Mitigation Measure	Frequency of Monitoring	Time Frame	Performance Indicator	Monitoring and Reporting	Cost (ZMK)
		hydrocarbons storage areas Contaminated soils to be disposed through licenced companies					
General waste management	- Land and water pollution - Animal-human conflict	Proper management of waste	Monthly	Long term	- Bins on site - No indiscriminate waste disposal - No signs of open air burning - No animals around waste storage areas	site Manager/Contractor	1000
Air quality	Increased dust and loss of good air quality	- Avoiding open air burning of waste - Proper disposal of waste	Monthly	Short term	- Well managed waste bins - Area dust free	Site Manager/ Contractor	1000
Creation of employment and skills development	Improved incomes	Locals favoured in terms of employment	On-going	Long term	- Number of locals employed - Less cases with the labour office	Site Manager /Contractor	2000

Aspect	Impact	Mitigation Measure	Frequency of Monitoring	Time Frame	Performance Indicator	Monitoring and Reporting	Cost (ZMK)
Fire Outbreak	Loss of vegetation and fauna	• Fire-fighting equipment and training Provided • Proper storage of explosives	Weekly	Medium term	• No cases of fire	Site manager	2000
Human-animal conflict	Injuries and loss of life	• All DNPW guidelines adhered to	Daily	Medium term	• No cases of attacks	Site manager	1000

8.0 DECOMMISSIONING AND CLOSURE PLAN

The closure of the mine will be considered at the end of the operational phase of the project. If decommissioning is inevitable then a decommissioning and closure plan will be followed.

8.1 Objectives

The target closure outcomes of the decommissioning plan are to:

- Restore as much as possible of the project site to a condition consistent with the predetermined post-closure site use objective;
- Ensure that the project site is left in a condition which poses an acceptable level of risk to public health and safety;
- Minimize or prevent environmental degradation by water, by ensuring the site is left in a condition that is chemically and physically stable; and
- Comply with national and environmental regulations.

The following is the procedure for decommissioning;

- a. Notify relevant authorities such as ZEMA, Chirundu District Council, and other of the intention to cease activities.
- b. Removal of dilapidated structures
- c. Levelling the area and revegetation
- d. Conducting baseline studies on relevant parameters and continual monitoring

The decommissioning exercise will take into account the implementation of the environmental management plan highlighted above. Summary of the decommissioning activities and associated costs are indicated in the table below.

Table 7: Decommission budget

Activity	Objective	cost
Notify all relevant authorities	To ensure all relevant authorities and concerned parties are aware of the closure of the project and decommission report submitted	40,000
Leveling the area and revegetation	To re-profile the sites to original form and protect the public from injury risks	20,000
Baseline survey, water and air	Compliance	5000

9.0 BIBLIOGRAPHY

1. Development in Protected Areas: Guideline for Planning and Management, IUCN Gland, Switzerland and Cambridge, UK
2. ZEMA (2011) Environmental Protection and Pollution Control Act No. 12, Government printers, Lusaka
3. Tychsen, J; Mukofu, C; Msimuko, J; Zimba, K; Chadukwa, C; Chibonga, M; Phiri, C; Simukali, M; Nguni, M; Mwenya, C; Chinyamuka, L; Sanga, C; Chuula, T; and Milimo, I (2018), "ASM Handbook for Zambia" Geological Survey of Denmark and Greenland (GEUS), Copenhagen, Denmark, 218 pp.
4. Central Statistics Office (2020), "Census of Population and Housing", government printers, Lusaka, Zambia.
5. Environment and Natural Resources vol. 4. FAO Corporate Document Repository (downloaded from the internet)
6. GRZ 1965 The Republic of Zambia Soil Map Compiled from the Information Supplied by the Soils Section, Research Branch, Department of Agriculture, Lusaka, Zambia
7. Government of the Republic of Zambia; S. I. No.28 of 1997, the Environmental Impact Assessment Regulations.
8. (ZAWA, 2008) Atlas of the National Parks of Zambia, ZAWA Chilanga
9. The Environmental Management (Amendment) Act (EMA) No. 8 of 2024 as read together with the EMA no. 12 of 2011
10. Environmental Council of Zambia (2000), "State of the Environment", Zambia.
11. IFC, 2007: Environmental, Health, and Safety Guidelines for Retail Petroleum Networks Laws of the Republic of Zambia (1995) Ed., Vol.13.
12. SI No. 28 of 1997 Environmental Impact Assessment regulations.

10.0 DECLARATION OF AUTHENTICITY OF REPORT CONTENTS

We trust that the information described in this EPB is adequate and satisfies the laws and regulations of Zambia regarding construction of a mine . We also trust that the information meets the Zambia Environmental Management Agency's requirements for approval. To the best of my knowledge, the content of this report is the work of the consulting team and any work done by others have been quoted and referenced accordingly



.....
DIRECTOR
Mulonga Mining Cooperative Society Limited

11.0 APPENDICES

11.1 Consent from the Chief



REPUBLIC OF ZAMBIA

HRH Chieftainess Sikoongo's Own Palace

Sikoongo Main & Lusitu
Sub-Court
P.O. Box 30
Chirundu

Your Ref:

Our Ref:

13th March, 2024

MULONGA MINING COOPERATIVE
SOCIETY LIMITED
CHIRUNDU DISTRICT

Dear Sir/madam:

RE: GRANT OF CONSENT TO CONDUCT COPPER ORE EXPLORATION
AND MINING TO MULONGA MINING COOPERATIVE SOCIETY LIMITED
CERTIFICATE NUMBER 42024018393 IN MANDALE VILLAGE OF
CHIEF SIKOONGO IN CHIRUNDU DISTRICT SOUTHERN PROVINCE
COVERING PART OF SIKOONGO CHIEFDOM

Reference is made to the above subject matter, over your letter dated 12th
March 2024 concerning the subject matter.

The Sikoongo Chieftainess is pleased to **grant consent** to your cooperative
Mulonga Mining Cooperative Society Limited to carry out the proposed
mineral exploration works Under Certificate Number 42024018393
within our Chieftainess's part of Chirundu District, within the
requirements of the Mines and Minerals Development Act, No.11 of 2015,
and other relevant legislation.

Your proposed mineral exploration works and eventual mining works should
give preference to the Sikoongo Chieftainess's subjects in terms of jobs and other
social-economic benefits.

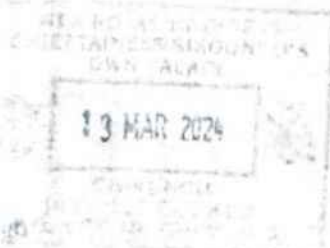
Failure to fulfillment of the agreement will lead to cancellation of this
consent.

Yours in Traditional Services

Karen Beatrice Kajembwe

HRH Chieftainess Sikoongo

CC: Traditional Council Sikoongo



Digitally signed by Thomas K. Rasmussen
 DN: cn=Thomas K. Rasmussen, o=IBM, ou=IBM
 Reason: I am the signer of this document.
 User-Agent: Mozilla/5.0 (X11; Linux i686; rv:1.9.0.12) Gecko/20100101 Firefox/3.6.12



Figure 2
Radiation 1

THE CO-OPERATIVE SOCIETIES REGULATIONS

I CERTIFY THAT, I have this day registered the **MULONGA MINING CO-OPERATIVE SOCIETY LIMITED** as a co-operative society under the Co-operative Societies Act.

and its postal address is: C/O SIANGWEEMU PRIMARY SCHOOL, P.O. BOX 21, CHIEF SIKOONGO AREA, MANDALE VILLAGE, CHIRUNDU, SOUTHERN PROVINCE, ZAMBIA

Hyde

Evans M. Kapekele
Deputy Registrar of Co-operatives

All correspondence must be addressed to Registrar



000-06578201-2014

11.3 Artisanal Mining Right

Licence Summary		Coordinates (IGCS, Arc1950)	
Code	37369-HQ-AMR	Latitude	Longitude
Type	Artisanal Mining Right	1. 28° 30' 00.00" S	28° 35' 12.00" E
Status	Application - Pending Doc	2. 28° 30' 00.00" S	28° 35' 24.00" E
Owner	Mulonga Mining Cooperative	3. 28° 30' 00.00" S	28° 35' 24.00" E
Application Date	09/03/2024 10:16:21	4. 28° 30' 00.00" S	28° 35' 12.00" E
Granted Date			
Expiry Date			
Commodities	Copper, Cobalt, Silver, Gold, Gemstones, Iron, Lead, Manganese, Mica, Nickel, Zinc		
Location	Southern Province		
Official Area	1,000 Ha		
Accounting Code			



11.4 Payment receipt

37369-HQ-AMR

REPUBLIC OF ZAMBIA - GENERAL RECEIPT N° G 1497431

N ORIGINAL
Accounts Form 40

STATION SK

Date 05-03-2024

RECEIVED from Mulonga Mining Co-op. Soc Ltd

The sum of: Kwacha Nine hundred only

in words K 900 --

For the credit of 12250

Details and references Payment of app for AmR



OFFICIAL STAMP

Designation APAd

For ink only. No erasures allowed.